① Qiskit program

```
4 circuit = QuantumCircuit(2)
5 circuit.h(0)
6 circuit.h(1)                fault
7 circuit.cx(0,1)
8 circuit.measure_all()
```

lines 10–16: run on a simulator, print

Expected $(|01\rangle+|10\rangle)/\sqrt{2}$; line 6 should apply X.

parse

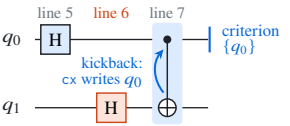
② srcML of line 7

```
<expr_stm pos:start="7:1"[d]>
 <expr><call>[a]
  <name><name>circuit</name>
   <operator>.</operator><name>cx[b]</name></name>
  <argument_list>(<argument><expr>
   <literal type="number">0[c]</literal>
   </expr></argument>,<argument><expr>
   <literal type="number">1</literal>
   </expr></argument>)</argument_list>
 </call></expr></expr_stmt>
```

a `call` → one operation **b** `cx` → gate
c `0`, `1` → q_0 control, q_1 target
d position → line 7 wire rule: writes both

interpret

③ Operation trace



shaded: kept by the wire rule's backward pass from q_0. The terminal measurement of line 8 is outside the criterion.

slice

④ Slice of q_0

wire PDG

```
4 circuit = QuantumCircuit(2)
5 circuit.h(0)
6 circuit.h(1)
7 circuit.cx(0,1)
8 circuit.measure_all()
```

wire rule: fault line 6 kept
classical PDG: line 5 only, fault missed
full square: quantum line; light: classical line (circuit construction).